Andrei Stefan Bejgu (he/him)

Date of Birth: 16-11-1995
Location: Rome, Italy
Languages: English (Fluent), Italian (Native), Romanian (Native)

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AI researcher with **9+ years of experience**. Today I am Senior AI Applied Scientist at Syllotips, working on how its agents plan, use tools, cite sources and improve over time. Before that I spent 8 years at Babelscape, growing from software engineer to NLP researcher, earned an industrial Ph.D. in AI at Sapienza and published at ACL, EMNLP, EACL and NAACL.

Experience

Syllotips
Remote

Senior AI Applied Scientist
Oct. 2025–Present

- Redesigned and rebuilt the **deep-research agent**, the main revenue product, with tool calling, shared working memory and a harness with tool budgets and bounded replanning, raising answer quality from 78% to 92%.
- Fine-tuned small open-weight LLMs (0.6B to 32B) on a **supercomputer** to replace three commercial LLM calls for answer confidence estimation with a single model.
- Designed the **MCP** layer that connects the agents to enterprise tools such as Atlassian, HubSpot and Intercom: one architecture for cloud and private cloud, OAuth handling, and tool results the agents can cite.
- Built a sandboxed **code interpreter** that lets agents analyse spreadsheets and produce reports and files, on cloud and private cloud alike.
- Rebuilt the **assistant agent** that helps experts close the loop on issues: it investigates what went wrong and proposes a solution for them to review.
- Built **self-hosted model serving** on vLLM across multi-GPU clusters: open-weight LLMs with up to 1M-token context, plus small decision models the agents use for fast routing and grading, so the agents can run fully on our own infrastructure.

Babelscape
Rome, Italy

NLP Researcher
Oct. 2021–Sep. 2025

- Led the development of **Vera**, a multilingual fact-checking product built on LLMs and RAG: real-time retrieval over web and proprietary data, claim extraction and verification.
- Fine-tuned open-source LLMs on Slurm clusters for text-to-SQL and claim verification, and deployed them for industry partners.
- Worked on **Minerva**, the first Italian LLM: supervised fine-tuning, vocabulary adaptation and continual training.
- Turned research into open models and datasets: **Word Sense Linking** (ACL 2024), **LLM-Oasis** (Computational Linguistics) and **CroCoAlign** (EACL 2024).
- Coordinated research and engineering work across teams and mentored interns from prototype to deployment.

NLP Engineer
Jun. 2020–Oct. 2021

- Built **Semantic Paths**, a multilingual semantic search engine for scientific papers.
- Built **Word Sense Disambiguation** models for large-scale multilingual text, combining embeddings, vector search and knowledge-based disambiguation.

Software Engineer
Jul. 2017–Jun. 2020

- Built **myKnowledgeGraph**, a knowledge graph extraction and visualization tool.
- Designed the company's **Java NLP pipeline** for multilingual text processing.
- Built concept identification and extraction for trademark names.

CINI - Consorzio Interuniversitario Nazionale per l'Informatica
Rome, Italy

Software Developer
Dec. 2016–Jan. 2017

Created an advanced tool for lexicon-semantic stylometric analysis, enabling efficient processing of large-scale texts.

Education

Ph.D. in Artificial Intelligence
Sapienza University of Rome Jan. 2025
Focus: lexical semantics for multilingual NLP, and word- and sentence-level representations for RAG and LLMs. Joint industrial Ph.D. with Babelscape. **Advisor:** Prof. R. Navigli

MSc in Computer Science
Sapienza University of Rome, ***Grade:*** *110/110 with honors* Nov. 2021
Thesis: Exploring language-agnostic latent spaces for sentences. **Advisor:** Prof. R. Navigli

BSc in Computer Science
Sapienza University of Rome, ***Grade:*** *107/110* Jun. 2019
Thesis: Knowledge-Based Word Sense Disambiguation leveraging synset embeddings. **Advisor:** Prof. R. Navigli

Publications

[1] A. Scirè*, **A. S. Bejgu***, S. Tedeschi, K. Ghonim, F. Martelli, and R. Navigli. "Truth or Mirage? Towards End-to-End Factuality Evaluation with LLM-OASIS". In: *Computational Linguistics* (2025).

[2] L. Moroni, T. Bonomo, L. Gioffré, L. Xu, D. Fedele, L. Colosi, **A. S. Bejgu**, A. Scirè, and R. Navigli. "What we Learned from Continually Training Minerva: a Case Study on Italian". In: *CLiC-it*. 2025.

[3] L. Moroni, G. Puccetti, P.-L. H. Cabot, **A. S. Bejgu**, A. Miaschi, E. Barba, F. dell'Orletta, A. Esuli, and R. Navigli. "Optimizing LLMs for Italian: Reducing Token Fertility and Enhancing Efficiency Through Vocabulary Adaptation". In: *Findings of NAACL*. 2025.

[4] K. Ghonim, **A. S. Bejgu**, and R. Navigli. "Conceptpedia: Leveraging Traditional Approaches for Multimodal Image-Text Representation". In: *EMNLP*. 2025.

[5] **A. S. Bejgu**, E. Barba, L. Procopio, A. Fernández-Castro, and R. Navigli. "Word Sense Linking: Disambiguating Outside the Sandbox". In: *ACL*. 2024.

[6] F. Molfese, **A. S. Bejgu**, S. Tedeschi, S. Conia, and R. Navigli. "CroCoAlign: A Cross-Lingual, Context-Aware and Fully-Neural Sentence Alignment System for Long Texts". In: *EACL*. 2024.

[7] F. Martelli, **A. S. Bejgu**, et al. "XL-WA: a Gold Evaluation Benchmark for Word Alignment in 14 Language Pairs". In: *CLiC-it*. 2023.

* equal contribution

Other Activities

Open Source: Word Sense Linking and ConceptPedia models and datasets on Hugging Face.
Reviewer Activity: Reviewer for EACL (2023), ACL (2023, external reviewer), EMNLP (2023, external reviewer), ICLR (2024, external reviewer), and ACL ARR (2024 and 2025).
Teaching Activity at Sapienza University of Rome:

- **Natural Language Processing Course (2022–2023)**: Designed and led practical lab sessions and assignments, for master's students.
- **Hot Topics in NLP (2023–2024)**: Designed and ran one of the three main course assignments.

Scholarships and Awards

- Best Employee Award, Babelscape 2021
- Merit Scholarship Winner, LazioDiSCo.it 2018–2020